User-Adaptive Verbalization of Explanations for Explainable AI

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Student Declaration

I hereby declare that the work presented in the report entitled **"User-Adaptive Verbalization of Explanations for Explainable AI"** submitted by me for the partial fulfilment of the requirements for the degree of *B.Tech. in Computer Science & Engineering* at Indraprastha Institute of Information Technology, Delhi, is an authentic record of my work carried out under the guidance of **Dr. Raghava Mutharaju** . Due acknowledgements have been given in the report for all material used. This work has not been submitted elsewhere for the reward of any other degree.

Anushka Korlapati, Vimal Jayant Subburaj **Place & Date:** December 7, 2025

Certificate

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Dr. Raghava Mutharaju **Place & Date:** December 7, 2025

Abstract

This work explores an ontology-grounded framework for generating user-adaptive explanations across multiple biomedical tasks. To achieve this, we combine a biomedical language model, local feature explanations, and biomedical ontologies to produce readable justifications tailored to different levels of domain expertise. In addition, the project investigates how steering vectors could be used to further control the style and complexity of these explanations, outlining this as a direction for future development.

Keywords: (Explainable AI, LIME, ontology-based explanations, knowledge graphs, user-adaptive explanations, beginner and expert users, steering vectors, clinical language models, natural language justifications, biomedical tasks.)

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Chapter 1

Introduction

1.1 Background

Deep learning has demonstrated remarkable performance across biomedical tasks such as disease classification, diagnostic assistance, and phenotyping. However, these models often function as opaque "black boxes," making it difficult to understand the reasoning behind their decisions. This opacity poses substantial risks in biomedical settings, where reliability, accountability, and trust are essential [11]. As highlighted by recent surveys, a major barrier to deploying machine learning in healthcare is the lack of transparent and trustworthy explanations that clinicians and patients can meaningfully interpret [1].

The rise of Explainable Artificial Intelligence (XAI) attempts to address these challenges by offering post-hoc interpretability methods. Prominent techniques such as LIME [11] and SHAP [7] attribute feature importance to model predictions. While these approaches provide valuable insights into model behaviour, research has consistently shown that they suffer from critical limitations: they generate explanations that are local but not globally consistent, lack semantic grounding, are sensitive to perturbations, and often fail to align with human cognitive expectations [8]. Furthermore, most current XAI methods focus exclusively on numerical or visual artefacts rather than producing coherent, domain-specific natural language explanations [1].

Recent research converges on the idea that combining symbolic Knowledge Representation and Reasoning (KRR) techniques with data-driven Machine Learning (ML) can yield hybrid intelligent systems with improved interpretability [3]. Symbolic structures such as ontologies encode hierarchies, biomedical semantics, and clinically validated conceptual relationships, enabling interpretability that is more aligned with human reasoning. As emphasized in [8], trustable XAI must go beyond surface-level feature importance and incorporate domain-specific knowledge to generate explanations that are verifiable, contextually meaningful, and actionable.

A second major challenge is that explanations must be *user-adaptive*. According to recent findings in adaptive XAI [12], the effectiveness of an explanation depends heavily on the user's expertise, cognitive load, and familiarity with domain concepts. A novice user often benefits from simpli-

fied, high-level descriptions, whereas experts require deeper, mechanistic explanations referencing biomedical ontology terms. Traditional XAI techniques, however, produce static and homogeneous explanations regardless of user background, limiting their practical utility.

Taken together, these insights highlight two significant gaps in biomedical XAI: (i) the absence of ontology-grounded semantic explanations, and (ii) the lack of user-adaptive natural language explanations. Bridging these gaps requires integrating XAI, biomedical ontological reasoning, and modern generative language models into a unified pipeline capable of producing meaningful, context-aware narratives tailored to the user.

1.2 Motivation

Biomedical datasets inherently contain domain semantics, yet classical XAI methods do not map feature-level explanations to clinically validated concepts. Ontologies such as UMLS, SNOMED-CT, and MeSH provide structured representations of biomedical knowledge, including hierarchical relations, synonyms, semantic types, and clinically meaningful groupings [4]. Leveraging these ontologies enables the transformation of raw feature importances into explanations that align with biomedical reasoning processes.

In parallel, modern large language models (LLMs) such as BioBART, Gemma, and GPT have demonstrated powerful abilities in natural language generation, contextual integration, and explanation synthesis. However, without explicit biomedical context, LLM-generated explanations can suffer from hallucination, lack of factual grounding, or oversimplification. The literature emphasizes that explainability must connect model behaviour to domain knowledge in a way that is both accurate and comprehensible [1, 8].

Furthermore, as highlighted in recent adaptive-explanation frameworks [12], explanations should be personalized to the recipient's level of expertise. A one-size-fits-all explanation risks overwhelming novice users or under-informing clinical experts. Therefore, the integration of ontological reasoning with user profiling and LLM-based generation constitutes a promising approach to producing explanations that are both semantically meaningful and adaptively tailored.

Thus, combining (i) feature-level XAI techniques, (ii) ontology-driven semantic enrichment, and (iii) user-adaptive, transformer-based natural language generation offers a path toward practical, trustworthy, and clinically relevant explainability in biomedical models.

1.3 Problem Definition

Biomedical machine learning models produce high-performing classification outputs, yet their decision-making processes remain opaque and insufficiently interpretable. Given an input sample x and its corresponding model prediction $f(x)$, existing techniques such as LIME and SHAP identify influential features but provide no structured biomedical meaning, no ontological contextualization, and no coherent natural-language explanation.

Moreover, these explanations remain static and undifferentiated across users regardless of their expertise, violating the principles of adaptive and trustable explanation design described in recent literature [12, 1]. In clinical scenarios, explanatory needs vary widely across medical professionals, researchers, and lay users, making fixed explanations impractical and potentially misleading.

Thus, the central problem addressed in this work is:

> *How can raw feature-based XAI outputs be transformed into ontology-grounded, context-aware, and user-specific natural language explanations suitable for different expertise levels in biomedical applications?*

Answering this requires a hybrid approach that integrates XAI, ontological reasoning, user modeling, and controlled natural language generation.

1.4 Proposed Approach

Our proposed pipeline (Figure 1.1) addresses this problem through a multi-stage, hybrid neuro-symbolic framework.

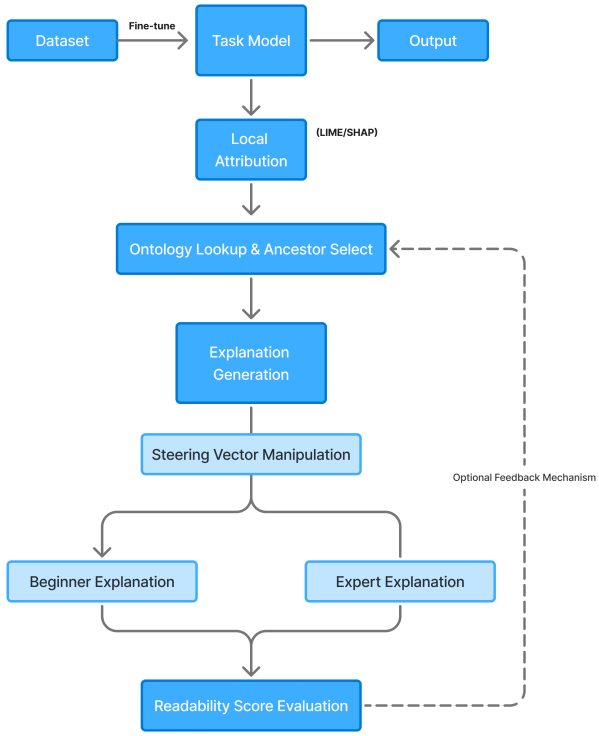

Figure 1.1: Proposed pipeline for user-adaptive XAI using ontology and natural language generation.

1. **Task Model Prediction**: A pretrained or fine-tuned biomedical classifier produces an output for a given input.

2. **Feature Importance Extraction**: LIME/SHAP generates feature attributions for the prediction.

3. **Ontology-Based Semantic Enrichment**: The identified important features are mapped to biomedical ontologies. Relevant ancestors, semantic definitions, hierarchical relations, and high-level concept groups are retrieved. This forms the first level of user adaptiveness, where high-level or low-level ontology ancestors are selected based on user type.

4. **User-Adaptive Explanation Generation**: The model output, input features, ontology-derived context, and user category are provided to a transformer-based explanation model (e.g., Gemma or BioBART). The model synthesizes natural language explanations tailored to beginner or expert profiles.

5. **(Future Work) Steering Vector Manipulation**: Steering vectors may be applied to dynamically adjust explanation complexity, controlling narrative detail, conceptual depth, or technicality.

This pipeline ensures semantically grounded, context-rich, and adaptively personalized explanations for diverse biomedical users.

Chapter 2

Literature Review

This chapter reviews existing works on healthcare knowledge graphs, ontology engineering, difficulty modelling, biomedical language models, explainable AI, steering vectors, and user-adaptive XAI, and positions the proposed framework within this landscape.

2.1 Knowledge Graphs in Biomedical Domain

Healthcare knowledge graphs provide the structural backbone for this project. Yang et al. survey HKGs as semantic graphs that capture domain entities such as diseases, drugs, genes, and phenotypes along with their relationships, and describe how they are built either from scratch or by integrating existing ontologies and clinical data sources. Their review distinguishes model-free uses (e.g., querying, visualization) from model-based uses in which HKGs enrich downstream machine learning models, concluding that HKGs are key enablers for clinical decision support and interpretable analytics. This motivates the choice to ground explanations in disease and symptom knowledge graphs rather than relying only on raw text features [18].

While Yang et al. argue for the importance of HKGs in general, Azim and Imran provide a practical perspective on how biomedical knowledge graphs are actually constructed. They outline a pipeline starting from curated databases and literature, moving through ontology-based data access and integration, and using NLP techniques such as named entity recognition and relation extraction to populate graphs. Their case studies demonstrate that well-designed biomedical KGs can integrate heterogeneous data and support tasks ranging from information retrieval to hypothesis generation, which aligns directly with this project's use of existing disease and symptom ontologies as an external source of medically grounded structure for explanation [2].

2.2 Ontology Engineering and Difficulty Modelling

Both of these works assume robust ontologies underlie the graphs, which leads naturally to ontology engineering guidelines. Noy and McGuinness describe practical steps for building ontologies,

including identifying key domain terms, deciding what should be modeled as a class versus an instance, designing class hierarchies, and defining properties and constraints. They stress choosing an appropriate level of granularity for the intended application and highlight how OWL reasoners can classify ontologies, infer new subclass relationships, and ensure logical consistency. These principles inform how disease and symptom ontologies are selected and used in this project, and why reasoners are employed to pull ancestors, related conditions, and definitions when constructing explanations [10].

Beyond representing knowledge, this work needs a way to relate ontological structure to user expertise. Vinu and Kumar propose a difficulty-level model for ontology-based factual questions that combines ontology-derived features with Item Response Theory. They quantify question difficulty using metrics such as popularity, specificity, selectivity, and coherence, and train separate logistic regression models for experts, intermediates, and beginners to predict difficulty labels. Their results show that this feature set predicts perceived difficulty more effectively than earlier methods, which directly inspires the approach in this project: ontology concepts and ancestors are scored using similar structural features and grouped into beginner-friendly or expert-level context [17].

2.3 Biomedical Language Models and Tasks

Complementing the symbolic layer, modern biomedical NLP models provide the base predictions that must be explained. Lee et al. introduce BioBERT, a domain-specific adaptation of BERT further pre-trained on large PubMed and PMC corpora to better capture biomedical terminology and syntax. Their experiments demonstrate consistent performance gains over general-domain BERT on several biomedical benchmarks, confirming that in-domain pretraining is crucial for biomedical tasks. This supports the use of biomedical transformers like BioBERT as the underlying prediction models in this project [6].

2.4 Explainable AI and Ontology-Grounded Explanations

Explainable AI methods link model predictions with interpretable features that can then be grounded in an ontology. Ribeiro, Singh, and Guestrin present LIME, a model-agnostic framework that fits local surrogate models around individual predictions to highlight the most influential features. Their case studies show how LIME helps detect data leakage, uncover spurious correlations, and improve transparency in text classification. In this project, LIME extracts key biomedical tokens and phrases from each prediction, which are subsequently mapped onto ontology entities and expanded using reasoner-derived relations [11].

2.5 Controllable and User-Adaptive Explanations

Recent advances in steering vectors suggest additional ways to control model behaviour and explanation style. Subramani et al. demonstrate that pretrained language models contain latent steering vectors—activation directions that can reconstruct or modify sentences when injected into model layers. Their findings show that steering applied to middle layers enables unsupervised attribute control. Although steering vectors are not yet implemented in this project, they motivate the planned extension for learning beginner- and expert-oriented explanation directions [13].

Finally, user-adaptive XAI research connects these tools to user needs. Work on user-centred explainability highlights that explanation effectiveness varies across users and that explainability should be aligned with user goals, mental models, and cognitive states. Studies show that demographics and personality traits have weak influence on how users interpret explanations, suggesting that coarse-grained groups (e.g., experts vs. lay users) may be more actionable. Broader surveys and frameworks highlight the importance of trust, transparency, and contextual adaptation in XAI system design [1, 8, 3, 12].

Our proposed framework builds directly on these insights by combining LIME-based feature extraction, ontology-driven reasoning, and difficulty-aware content selection to generate explanations tailored for both beginners and experts today, while leaving room for steering-based fine control in future work.

Chapter 3

Methodology

3.1 Overview of the User–Adaptive XAI Pipeline

The proposed system follows a task-agnostic, modular pipeline that transforms raw biomedical text inputs into user-adaptive natural language explanations. The entire workflow consists of four interconnected modules:

1. **Task Model Prediction**: A fine-tuned biomedical model produces a prediction for the given input.
2. **Feature Importance Extraction**: Local feature attribution is computed using LIME to identify salient tokens.
3. **Ontology-Based Semantic Enrichment**: Salient tokens are grounded in biomedical ontologies, and difficulty-aware ancestor selection is performed.
4. **User-Adaptive Explanation Generation**: A transformer-based LLM produces tailored explanations for beginner or expert users.

This pipeline is designed to generalize across several biomedical NLP tasks, including NER, classification, relation extraction, and QA. Each module is elaborated in the following sections.

Current Implementation Scope

Although the proposed framework is designed to be task-agnostic, the present prototype executes each biomedical NLP task through its **own isolated pipeline instance**. That is, for a given task such as multi-class disease classification, NER, relation extraction, or sentence classification, we instantiate the full four-stage pipeline independently:

- loading a task-specific fine-tuned BioBERT model,
- applying LIME to that model's predictions,

- grounding the extracted features in an ontology chosen for that specific task,
- generating user-adaptive explanations.

Thus, the current system should be understood as a **family of parallel pipelines**, one per task rather than a single unified multi-task explanation engine.

In future work, we aim to consolidate these independent pipelines into a **single universal explanation model** capable of consuming arbitrary task outputs (NER spans, classification labels, relations, QA answers) and producing user-adaptive explanations through a unified interface. This is enabled by the modular nature of the present design.

3.2 Task Models and Datasets

This stage performs the underlying biomedical NLP task. Although the overall explainability pipeline is task-agnostic, each task requires its own fine-tuned BioBERT model and task-specific dataset. At present, these tasks are implemented *independently*: each task has its own end-to-end explanation pipeline. A unified cross-task explanation system will be developed in future work.

The pipeline currently supports:

- Multi-class Medical Abstract Classification,
- Medical Named Entity Recognition (Disease NER),
- Biomedical Question Answering.

Table 3.1 summarises the datasets, task types, and ontologies used.

Table 3.1: Biomedical tasks, datasets, models, and ontologies used in the pipeline

Dataset	**Task Type**	**Finetuned Base Model**	**Ontology / KG Used**
Medical Abstracts Text Classification Dataset	Multi-class Medical Abstract Classification	BioBERT	MeSH, DOID
NCBI Disease Corpus	Disease Named Entity Recognition	BioBERT	MeSH, UMLS
BioASQ (subset)	Biomedical Question Answering / Sentence Classification	BioBERT	MeSH
SymbiPredict	Disease Classification	BioBERT	MeSH,DOID,SYMP

These task-specific models act as black-box predictors, and their predictions are later explained through LIME, ontology grounding, and the user-adaptive explanation model.

3.2.1 Task Descriptions and Dataset Details

Below, we describe each task used in our experiments. Each task currently operates with its own independent pipeline instance.

Multi-Class Medical Abstract Classification

This task involves assigning a biomedical abstract to one of five high-level disease-related categories. We use the **Medical Abstracts Text Classification Dataset**, a curated collection of 11,550 medical abstracts spanning five major conditions. The dataset provides a realistic distribution of biomedical topics, with class frequencies summarised below:

- Neoplasms: 3163 abstracts
- Digestive system diseases: 1494 abstracts
- Nervous system diseases: 1925 abstracts
- Cardiovascular diseases: 3051 abstracts
- General pathological conditions: 4805 abstracts

The goal is to assign each abstract to exactly one disease category. This task is well-suited for evaluating our explanation pipeline, as biomedical abstracts contain rich domain-specific terminology and hierarchical medical concepts.

For ontology grounding, we use the **Disease Ontology (DOID)** and **MeSH (Medical Subject Headings)**. The rationale for this choice is:

- Both ontologies contain well-defined disease hierarchies aligned with the five categories in the dataset.
- DOID provides explicit parent–child disease relations useful for ancestor selection.
- MeSH includes synonyms, preferred labels, and hierarchical tree codes, facilitating robust ontology matching.
- Using only disease-focused ontologies ensures semantically consistent grounding while avoiding noise from broader ontologies.

These ontologies collectively provide high-quality semantic structures that support both shallow (beginner-friendly) and deep (expert-oriented) biomedical concept explanations.

Biomedical Question Answering (BioASQ Subset)

This task focuses on answering biomedical questions using a subset of the BioASQ Task 6b dataset, specifically the `test-factoid-6b-1.json` test set containing 121 question-answer pairs across multiple biomedical articles and paragraphs.

The task is formulated as extractive question answering, where the fine-tuned **BioBERT** model (`dmis-lab/biobert-base-cased-v1.1-squad`) identifies the most probable answer span from given context passages. Representative examples include:

- "What does davunetide do to microtubules?" → "impairments in axonal transport"]
- "What is a coligo?" → "L-loop"

For ontology grounding, we use the **MeSH (Medical Subject Headings)** ontology. The rationale for this choice is:

- Comprehensive coverage of biomedical terminology relevant to question answering contexts
- Hierarchical structures supporting multi-level explanations (beginner/intermediate/expert)
- Rich concept definitions and synonyms enabling precise term grounding
- Standardized biomedical vocabulary ensuring semantic consistency across diverse QA topics

MeSH grounding supports the full explanation pipeline by linking model predictions to standardized biomedical concepts with hierarchical relations, enabling both shallow (beginner-friendly) and deep (expert-oriented) explanations through OwlReady2-based ancestor selection.

3.3 Feature Importance Extraction using LIME (Module 2)

Once the predictive model outputs a label or entity span, the next step is to identify which input features influenced the decision. Local Interpretable Model-Agnostic Explanations (LIME) is used for this purpose.

LIME performs the following steps:

- tokenizes the input text,
- generates perturbed versions of the text by masking or modifying tokens,
- observes how these perturbations affect model predictions,
- fits a sparse linear surrogate model,
- surfaces the most influential tokens for the prediction.

These high-weight tokens serve as anchors for ontology lookup. The rationale for using LIME is that it provides explicit token-level contributions that are directly mappable to biomedical ontologies. Implementation details appear in Notebook Section 3.

3.4 Ontology-Based Semantic Enrichment (Module 3)

The salient terms identified by LIME are enriched with biomedical semantics. This enrichment step consists of:

- case normalization and lemmatization,
- synonym expansion (ontology labels),
- ID-based and string-based concept lookup in MeSH, DOID, UMLS, CTD,
- retrieving:
  - preferred names,
  - hierarchical ancestors,
  - semantic relations,
  - disease–symptom or disease–chemical links,
  - high-level ontological groupings.

This forms a structured interpretation layer that captures biomedical meaning beyond raw tokens.

3.4.1 Ontology Selection Strategy

Biomedical ontologies can be selected dynamically at inference time using federated services such as the **BioPortal API**, which provides programmatic access to ontology classes, synonyms, and mappings across hundreds of biomedical knowledge sources. However, to ensure reproducibility and computational stability, our current prototype adopts a simplified strategy: for each task, we manually select the most relevant ontology (e.g., MeSH or DOID for disease classification, CTD for chemical–disease relations, UMLS for NER). These ontologies are downloaded locally and loaded through `owlready2` for fast lookup and ancestor extraction.

This controlled setup allows consistent scoring, depth computation, and stability in the user-adaptive difficulty model, while retaining compatibility with future integration of real-time BioPortal querying in a unified multi-task setting.

3.4.2 Extracting Hierarchical Ancestors

For each mapped concept, we extract a chain of ancestors (general → specific). However, not all ancestors are equally interpretable, motivating the need for difficulty modeling.

3.5 Difficulty Modeling using Ontologies (Module 3b)

This stage forms the **first level of user adaptiveness**. We adopt the difficulty model of Venugopal and Kumar [16], which treats interpretability as a function of:

- **Specificity**: how deep or technical a concept is,
- **Popularity**: how frequently the concept appears in ontology relations.

3.5.1 Specificity

Defined as:

$$S(c) = \frac{\text{depth}(c)}{\text{max_depth}}$$

Higher specificity implies deeper and more technical concepts. This aligns with findings from [16] that deeper nodes impose greater cognitive load.

3.5.2 Popularity

Defined as:

$$P(c) = \frac{\sum_{r \in R} I(c \in \text{range}(r) \cup \text{domain}(r))}{|C|}$$

More popular concepts tend to be easier for non-experts.

3.5.3 Score Formulation

We assign difficulty-aware scores as:

$$\text{score}_{\text{beginner}}(c) = 0.6P(c) + 0.4(1 - S(c))$$

$$\text{score}_{\text{expert}}(c) = S(c)$$

$$\text{score}_{\text{intermediate}}(c) = 1 - |S(c) - 0.5|$$

These reflect cognitive preferences for different users:

- Beginners prefer broad, familiar concepts.
- Experts prefer deep, technical concepts.
- Intermediate users prefer mid-level concepts.

This scoring determines which ontology ancestors are passed to the explanation model.

3.6 User-Adaptive Explanation Generation (Module 4)

The final stage synthesizes all information into a user-tailored explanation using a transformer-based LLM such as Gemma or BioBART. The LLM receives a structured representation consisting of:

- model prediction,
- salient LIME tokens,
- mapped ontology concepts,
- selected user-adaptive ancestors,
- semantic relations,
- an explicit user category flag (Beginner/Expert/Intermediate).

No finetuning is required; user adaptiveness is controlled through templated prompts. Beginner prompts emphasize clarity and generality; expert prompts preserve medical precision.

This prompt-based formulation enables seamless integration of future steering vector mechanisms for explanation style control.

3.7 Readability and Faithfulness Evaluation (Module 5)

Explanation quality is evaluated along two axes: **readability** and **faithfulness**.

3.7.1 Readability Metrics

To measure linguistic complexity across different user modes, we compute:

- Flesch Reading Ease (FRE),
- Flesch–Kincaid Grade Level

These metrics represent lexical difficulty, syntactic complexity, morphology, and sentence structure.

3.7.2 Faithfulness Metrics

Faithfulness captures how well the explanation aligns with true model behavior and ontology facts. We use:

1. **LIME Coverage**: The proportion of high-weight LIME tokens referenced directly or indirectly in the explanation.

2. **Ontology Support**: Verifies that biomedical claims in the explanation are grounded in MeSH, DOID, CTD, or UMLS.

We justify the use of LIME based on established validation benchmarks from prior work: LIME demonstrates high recall of interpretable model features, improves user trust judgment (F1 score), aids model selection, and reveals spurious correlations (e.g., the wolf–husky example).

These metrics provide a principled starting point before integrating steering vector–based control of explanation style.

Chapter 4

Results

This chapter presents the results obtained by applying our user-adaptive XAI framework across multiple biomedical NLP tasks. For each task, we report both model performance (where applicable) and explanation-level evaluation using readability metrics and ontology-grounded faithfulness checks. All results correspond to the pipeline described in Chapter 3.

4.1 Task: Multi-Label Text Classification

This section presents the explanation-level evaluation conducted on the multi-label biomedical text classification pipeline. Unlike traditional model evaluation, our focus here is on analysing how explanation complexity varies across user categories and ontology configurations.

4.1.1 Experimental Setup

A complete explanation pipeline was constructed for the multi-label classification task, consisting of:

1. a finetuned BioBERT classifier,
2. LIME-based feature importance extraction,
3. ontology-grounded concept mapping,
4. user-adaptive ancestor selection (when enabled),
5. transformer-based explanation generation.

To evaluate the system, **50 random samples** were selected from the dataset. Each sample was passed through the pipeline under four conditions:

- **Beginner (Normal Mode)** — user-adaptive ancestor selection enabled

- **Expert (Normal Mode)** — user-adaptive ancestor selection enabled
- **Beginner (Full Mode)** — all ontology ancestors included
- **Expert (Full Mode)** — all ontology ancestors included

This allowed us to disentangle:

1. the effect of user category,
2. the effect of ontology depth,
3. the interaction between the two.

Readability metrics were computed for every generated explanation and averaged across samples.

4.1.2 Readability Comparison Across User Types

Readability was quantified using:

- **Flesch Reading Ease (FRE)** — higher = simpler text
- **Flesch–Kincaid Grade Level (FKGL)** — higher = more complex text

Normal Mode (Adaptive Ancestor Selection)

Table 4.1 summarises readability differences when only the top-k ontology ancestors were used.

User Category	FRE (↑)	FKGL (↓)
Beginner	31.23	12.04
Expert	8.60	16.78

Table 4.1: Readability metrics for Normal Mode (top-k ancestor selection).

A clear separation appears between user categories:

- Beginner explanations are substantially simpler.
- Expert explanations are more technical and require a higher reading level.

This confirms that the difficulty-aware ancestor selection mechanism effectively modulates explanation complexity.

Conclusion (Normal Mode). The readability gap demonstrates that the model successfully tailors explanations according to user category when ontology depth selection is enabled.

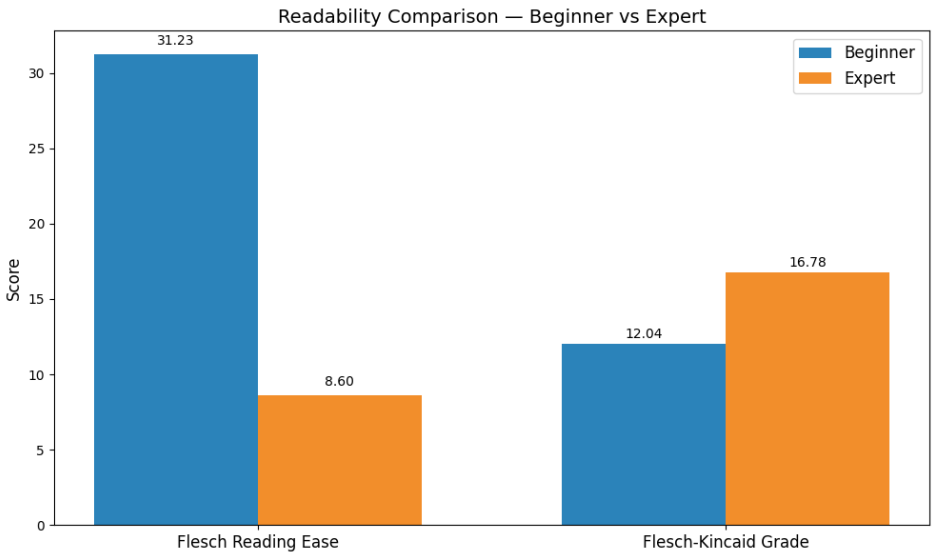

Figure 4.1: Beginner vs. Expert readability comparison in Normal Mode.

4.1.3 Ablation: Ontology Depth Control

To isolate the role of ontology depth, ancestor selection was disabled and the LLM was provided with *all* ancestors associated with each biomedical concept.

Full Mode (All Ontology Ancestors)

Table 4.2 shows the readability metrics.

User Category	FRE (↑)	FKGL (↓)
Beginner	14.31	16.15
Expert	12.84	16.00

Table 4.2: Readability metrics for Full Mode (all ontology ancestors).

Unlike Normal Mode, the two user categories produce very similar readability scores. This occurs because:

- deeper and more numerous ancestors dominate the prompt,
- the LLM is forced to use technical biomedical language,
- user-adaptive cues become overshadowed by dense ontology content.

Conclusion (Full Mode). Providing the full ancestor list removes user adaptiveness. Both beginner and expert explanations become equally technical, highlighting that **ancestor selection is essential** for controlling explanation difficulty.

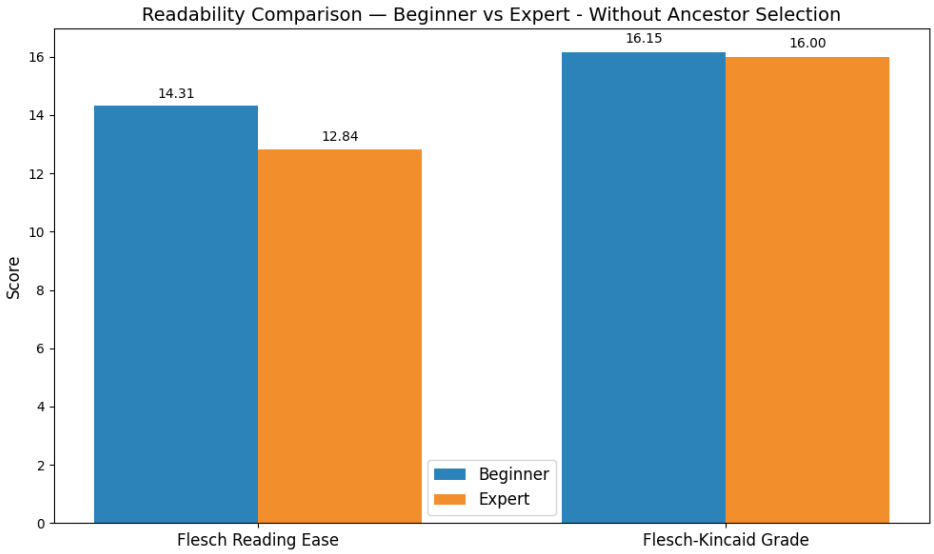

Figure 4.2: Readability comparison in Full Mode (all ontology ancestors).

4.1.4 Overall Findings

Across both evaluation conditions, we observe that:

- **User adaptiveness emerges only when ontology selection is active.**
- **Ontology depth directly influences linguistic complexity.**
- **Full Mode overwhelms the system with technical terms**, resulting in uniformly complex explanations.

These findings validate the central role of the difficulty-aware ontology scoring mechanism in shaping readable and user-appropriate explanations.

Future studies will incorporate additional metrics such as faithfulness scores, cross-task comparison, qualitative examples, and baseline comparisons.

Summary

This example demonstrates how the system identifies the important lexical cues responsible for a classification decision, maps them to biomedical ontologies, and synthesizes a coherent explanation grounded in these concepts. As we further integrate difficulty-aware context selection, this task will also support adaptive explanation generation across user profiles.

4.2 Task: Biomedical Question Answering (BioASQ Subset)

This section evaluates explanation behaviour for a biomedical question answering (QA) system applied to a subset of the BioASQ 6b benchmark. As in the previous task, our primary focus is not

model accuracy but the behaviour of the explanation-generation pipeline across user categories and ontology configurations.

4.2.1 Experimental Setup

The QA system was evaluated on the `test-factoid-6b-1.json` set, which contains 121 factoid-style biomedical questions. A finetuned **BioBERT QA model** (`dmis-lab/biobert-base-cased-v1.1-squad`) was used to extract answer spans from the provided context passages. To support structured explanation generation, predicted answers and influential context terms were grounded in the **MeSH** biomedical ontology.

For each QA sample, explanations were generated under four settings:

- **Beginner (Plain)** — LLM receives only question, answer, and context.
- **Expert (Plain)** — same as above, but phrased for experts.
- **Beginner (Ontology)** — MeSH concepts and ancestor definitions included.
- **Expert (Ontology)** — ontology-aware expert explanation.

Two representative BioASQ samples were used for controlled comparison, and readability metrics were averaged across them.

4.2.2 Readability Comparison Across User Types

Readability was quantified using:

- **Flesch Reading Ease (FRE)** — higher = simpler text
- **Flesch–Kincaid Grade Level (FKGL)** — higher = more complex text

Explanations Without Ontology

Table 4.3 summarises readability when explanations were generated without ontology information.

User Category	FRE (↑)	FKGL (↓)
Beginner	72.60	6.03
Expert	-1.85	18.66

Table 4.3: Readability metrics for QA explanations without ontology grounding.

A strong separation emerges:

- Beginner explanations are highly readable and accessible.

- Expert explanations become extremely technical (negative FRE indicates dense biomedical phrasing).

This shows that even without ontology, the prompting mechanism induces strong user adaptiveness.

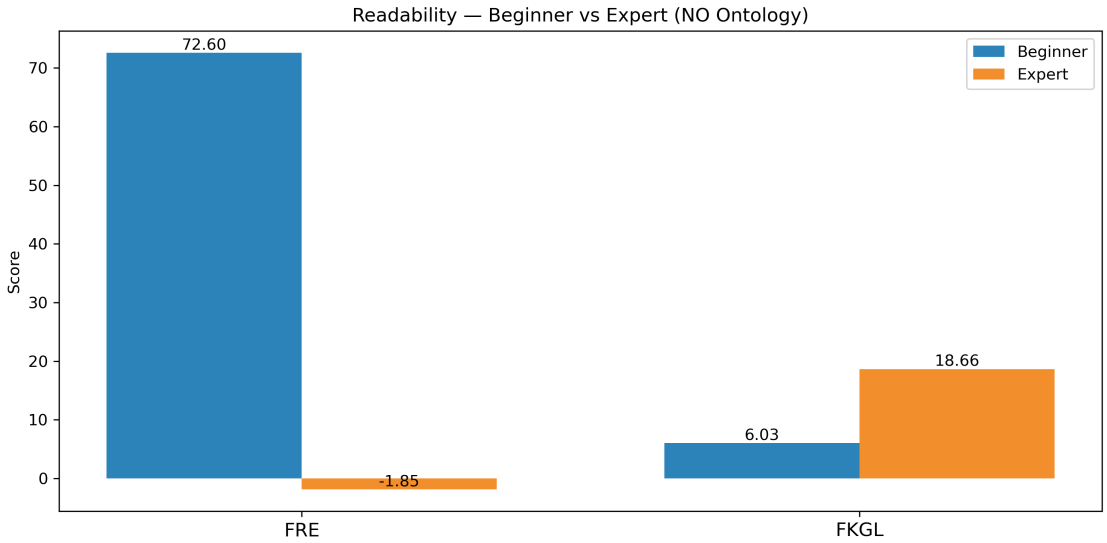

Figure 4.3: Beginner vs. Expert readability comparison for QA (no ontology).

Conclusion (No Ontology). Expert explanations are significantly more complex than beginner explanations, but the difference is unregulated, leading to extremely technical expert outputs.

Ontology-Grounded Explanations

Table 4.4 shows readability when MeSH concepts and ancestors were included.

User Category	FRE (↑)	FKGL (↓)
Beginner	39.85	12.85
Expert	12.06	17.26

Table 4.4: Readability metrics for QA explanations with ontology grounding.

Key observations include:

- Both beginner and expert explanations become more complex than in the plain setting.
- The readability gap narrows once ontology content is introduced.
- Ontology definitions introduce unavoidable domain-specific terminology that raises baseline difficulty.

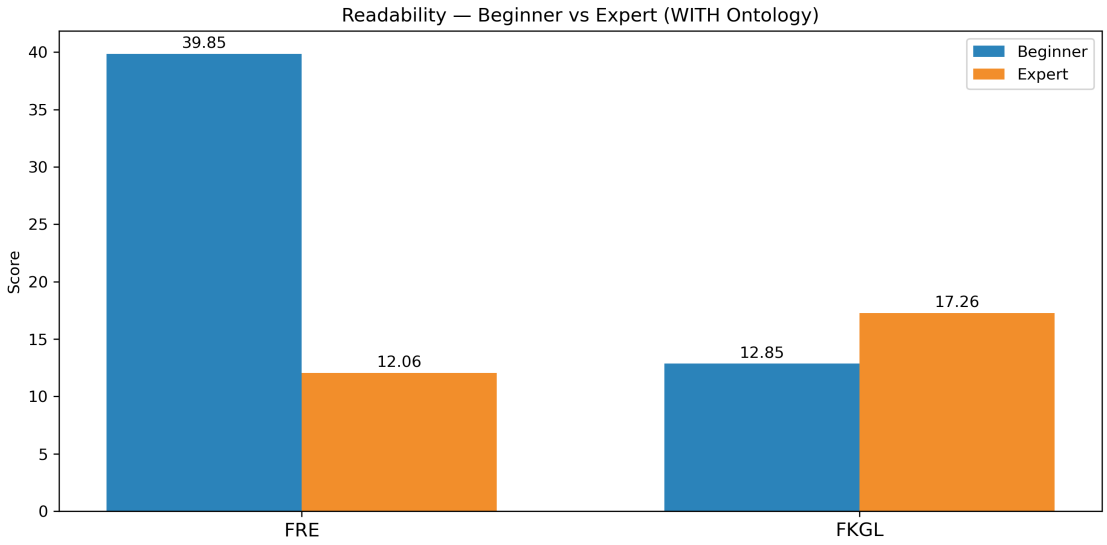

Figure 4.4: Readability comparison for QA explanations with ontology grounding.

Conclusion (Ontology). Ontology grounding introduces consistent biomedical terminology across user types, raising the overall complexity of explanations while still preserving distinctions between beginner and expert phrasing.

4.2.3 Overall Findings

Across both ontology conditions, we observe:

- **Strong user adaptiveness** emerges in both plain and ontology-grounded modes.
- **Ontology grounding increases explanation complexity**, especially for beginners.
- **Expert explanations remain consistently more technical**, but the gap narrows when ontology content is included.
- **Ontology controls and stabilises explanation structure**, preventing extreme divergence in difficulty.

Summary

This task demonstrates that the user-adaptive explanation framework generalises beyond classification to span-based biomedical reasoning. Ontology grounding plays a crucial role in shaping explanation complexity and ensuring semantic fidelity, while difficulty-aware prompting maintains appropriate stylistic differences between user categories.

Chapter 5

Future Works

This chapter outlines the major future directions for extending, unifying, and strengthening the proposed User–Adaptive XAI framework. While the current prototype demonstrates the feasibility of ontology-grounded and user-adaptive biomedical explanations, several key research directions remain to be explored.

5.1 Expanding to Additional Biomedical NLP Tasks

So far, the pipeline has been implemented for two tasks:

- Disease Named Entity Recognition (Disease NER),
- Multi-class Medical Abstract Classification.

To fully establish generality, we aim to extend the system to a wider range of biomedical NLP tasks such as:

- disease–symptom and chemical–disease relation extraction,
- medical summarisation,
- patient-triage decision explanation,
- multi-label clinical note classification,
- biomedical claim verification and evidence generation.

Each of these tasks brings different forms of model reasoning (token-level, sentence-level, pair-level), which will further validate the flexibility and robustness of our ontology-driven explanation pipeline.

5.2 Robust and Dynamic Ontology Selection

In the current system, we manually select a single, well-suited ontology for each task (e.g., DOID/MeSH for disease classification). While this ensures highly accurate concept grounding, it limits scalability and breadth.

Future work will focus on:

- **Automatic ontology selection**: dynamically determining the most relevant ontology (or multiple ontologies) for any given entity.
- **Multi-ontology fusion**: retrieving ancestor concepts from several biomedical ontologies simultaneously (MeSH, UMLS, DOID, SNOMED CT, CTD).
- **Semantic consistency scoring**: ranking ontologies by coverage, label quality, synonym richness, and hierarchical depth.
- **BioPortal-based lookup**: using the BioPortal API for live concept search, with fallbacks to local ontology stores when API access is unavailable.

A more robust ontology-selection mechanism will prevent failures due to missing concepts and allow the system to generalise to previously unseen biomedical terminology.

5.3 Unifying the XAI Model Across All Tasks

At present, each task uses its own task-specific XAI pipeline instance (model → LIME → ontology → explanation). A major goal is to **merge these into a single universal explanation engine** that can:

1. accept model predictions from *any* biomedical task,
2. extract the relevant tokens/entities,
3. perform ontology grounding automatically,
4. generate customised explanations for any user profile.

This unified design will reduce redundancy, simplify deployment, and align the system with real-world clinical decision-support systems where multiple NLP modules operate simultaneously.

5.4 Steering Vectors for User-Adaptive Explanation Control

A key future extension of our framework is the integration of **steering vectors** to achieve fine-grained, continuous control over explanation complexity. Steering vectors—also known as latent

directions, activation edits, or representation shifts—have recently emerged as a powerful method for modulating model behaviour without any retraining.

Recent work has demonstrated that hidden-state manipulation enables controllable reasoning [15], attribute modulation [14], and even systematic rewriting of model representations [5]. These approaches show that a transformer's internal activations encode semantically meaningful directions that can be added or subtracted to obtain predictable behavioural changes.

5.4.1 What Steering Vectors Are

A steering vector is typically computed by contrasting two sets of representations. For example, if H_{expert} and H_{beginner} represent hidden activations for expert-style and beginner-style explanations, a steering direction can be derived as:

$$v_{\text{expert}} = \text{mean}(H_{\text{expert}}) - \text{mean}(H_{\text{beginner}})$$

During inference, this vector is added to the model's internal states:

$$h' = h + \alpha \cdot v_{\text{expert}}$$

where α controls the strength of the effect. This approach draws on prior mechanistic interpretability work showing that latent directions correlate with abstract concepts, behaviours, and stylistic attributes [9].

5.4.2 Why Steering Vectors Fit Our Explanation Framework

Our system naturally produces paired explanation datasets:

- **Beginner explanations** — broad biomedical ancestors and simpler phrasing,
- **Expert explanations** — deeper ontology ancestors and denser terminology.

Such pairs are ideal for constructing steering vectors in the style of [14, 5]. This allows us to build latent "explanation complexity directions" that can shift the explanation model's behaviour along a continuous spectrum from beginner to expert.

5.4.3 Advantages for Adaptive XAI

Integrating steering vectors brings several benefits:

- **Continuous adaptiveness** rather than discrete mode switching.
- **Task-agnostic control**: once learned, vectors generalise across tasks.
- **No retraining needed**: the model's weights remain unchanged.

- **Manipulation of multiple stylistic factors**—terminology density, abstraction depth, sentence complexity.
- **Highly interpretable**: changes correspond to interpretable latent dimensions.

This makes steering vectors the most suitable mechanism for implementing personalised user-adaptive biomedical explanations.

5.5 Summary

Overall, the future work focuses on three major directions:

1. broadening task coverage,
2. strengthening ontology selection and concept grounding,
3. and developing a unified explanation engine enhanced with steering-based user adaptation.

Together, these extensions will complete the vision of a robust, user-controllable XAI framework capable of supporting real-world biomedical decision-making.

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